ZAMBIA SUGAR PLC
P.O. BOX 670240
ENGINEERING SPECIFICATION

PR (Contract) Nº.	CEF-NK-	
Department	Business Improvement	
Title	Proposed Upgrade of Nakambala Private School	
Nominee for Queries	Kelvin Mwila	
Project Manager	Mark Chewe	
Approved by:	Mulenga Chibwe	

PROJECT INTRODUCTION

The works comprise the construction of the Proposed Upgrade to Nakambala Private School including all associated civil, structural, plumbing, and electrical works. The Contractor shall carry out all works in accordance with approved drawings, specifications, and the instructions of the Project Manager.

SCOPE OF WORK

The Scope of Work will involve the following:

GENERAL PREAMBLES

- All works must comply with **Zambia Sugar PLC construction standards**, **NCC**, **ZEMA regulations**, and **ISO safety requirements**.
- The contractor must provide a **detailed schedule (Gantt chart) and work method statements** before execution.
- Quality assurance inspections must be conducted at **each critical phase of construction**.

PRELIMINARY & GENERAL ITEMS

- Site establishment, temporary fencing, and worker welfare facilities.
- Provision of **temporary utilities (power, water, sanitation)** during construction.
- Health and Safety (SHEQ) compliance, including **permits, fall protection, and confined space safety**.

CIVIL AND STRUCTURAL WORKS

- Site clearing and setting out shall be performed to approved benchmarks and coordinates.
- Excavations shall be to required depths and compacted in 150mm layers to 95% (Maximum Dry Density) MDD.

- Concrete works: Class 25MPa minimum for structural elements, mixed per approved design mix.
- Reinforcement: High yield steel bars to BS4449, minimum cover 25mm or as per drawings.
- Blockwork: Solid concrete blocks, minimum strength 7.5N/mm², laid in 1:4 cement-sand mortar.
- Provide DPC 1000 gauge and anti-termite treatment under all ground slabs and walls

ROOFING WORKS

- Roof structure shall comprise steel trusses or timber trusses as detailed.
- Roofing sheets: IBR profile, 0.58mm gauge, galvanized and pre-painted.
- Ridge capping and flashings in matching material.
- Gutters and downpipes: 150mm half-round PVC or equivalent approved..

FINISHES

- Plaster: 1:4 cement-sand mix, finished smooth.
- Floor finish: Cement screed or ceramic tiles per room schedule.
- Paint: Acrylic emulsion for walls, gloss for wood/metal surfaces, min. 2 coats over primer.
- Ceilings: PVC or gypsum board suspended ceiling, joints neatly finished.
- Doors/Windows: Hardwood or aluminium as specified; all ironmongery to be stainless steel or brass.
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PLUMBING AND DRAINAGE

- Cold water supply: uPVC/PPR pipes PN10 rating, properly jointed and tested.
- Sanitary fittings: White Vitreous China; approved brands.
- Drainage: uPVC pipes with inspection chambers at changes of direction.
- All works to comply with relevant water and sanitation regulations.

ELECTRICAL WORKS

- Electrical wiring: PVC insulated copper cables in concealed conduits.
- Switches and sockets: MK or approved equivalent.
- Lighting fixtures: LED fittings per layout drawings.
- Testing and commissioning per IEE Regulations (BS7671)

SITE WORKS AND EXTERNAL DEVELOPMENT

- Construct access roads, walkways, and parking areas using approved aggregates and paving.
- Erect boundary fencing and gates as per drawing details.
- Provide surface water drainage channels and landscaping to approved design

HEALTH, SAFETY & ENVIRONMENTAL (HSE)

- Contractor shall maintain a safe working environment in compliance with OSHA and ZEMA standards.

- Provide PPE for all workers and ensure daily toolbox talks are conducted.
- Control noise, dust, and waste to minimize environmental impact.
- Maintain proper waste segregation and disposal facilities on-site.
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TESTING, INSPECTION AND HANDOVER

- All materials and workmanship shall be subject to inspection and testing by the Engineer.
- Defective works shall be rectified at the Contractor's cost.
- Upon completion, provide as-built drawings, test certificates, and operation manuals.
- A defects liability period of 12 months shall apply from the date of handover.

PROJECT DOCUMENTATION DELIVERABLES

- Reinforcement shop drawings for suspended concrete roof.
- Steel fabrication & connection details before installation.
- Cable trench layout drawings for electrical routing.
- Final inspection & commissioning report upon completion.

INSTRUCTIONS TO CONTRACTORS

The duties of the Contractor during this phase of the project will include, but will not necessarily be restricted to, the following:

Provide schedule of works.

- Coordinating own labour to ensure the works are carried out in accordance with contract specifications and in the best interests of the Client with minimal disruption to existing operations.
- Monitoring the progress of the works against the agreed schedule and initiating such actions as are required to achieve the target completion deadlines.
- Controlling quality and quantity of all site materials under the terms of the contract; monitoring compliance with environmental, health and safety regulations, as well as the Client's SHEQ requirements.
- Issuing interim and final certificates and approvals for payment by the Client.
- Monitoring all corrective work done by, or on behalf of, the contractors during the contract defect liability periods; and,
- Performing any other services specifically related to the project which may be agreed between the Contractor and the Client, including the supervision or execution of any specified tests, assisting with the commissioning of works and the verification of performance guarantees.
- Necessary work permits are obtained timeously, including excavation permit.
- Dedicated safety watch to be on site at all times whilst construction is underway.
- The Contractor is responsible to review each General Arrangement (GA) drawing, reviewing the position of the construction on site and identifying any potential clashes with existing installed services prior to commencing with the Installation.
- Should any changes be deemed necessary the Contractor shall agree these changes with the Engineer on site and "Green Line" the GA drawing accordingly, which shall be signed by the engineer prior to installation commencing.
- Area to be always left neat and tidy.
- All work to be always supervised.
- All workers assigned to undertake this task should ensure that the correct PPE is always used as prompted by area and activity.
- Contractor to provide all material for any re-work due to poor workmanship (always consult at any critical stage).
- Clear off all debris and dispose of it to the designated dump site.

Material / Equipment to be supplied by Zambia Sugar PLC during construction:

- Contractor to supply own material / equipment.

Materials to be supplied by Contractor:

- Setting out equipment
- Excavation equipment
- Hand tools and equipment
- Storage facilities for their Tools/Equipment on site

CONSTRUCTION PROGRAMME

The Contractor shall submit to the Engineer, for approval, a copy of a Level three (3) programme conforming to the Engineer's requirements, showing the order of procedure and method in which it proposes to carry out the Works.

This programme is referred to herein as **"the Programme"** and shall be in the form of an activity linked Gantt chart subdivided into areas and activities to be performed by the Contractor.

The Programme shall be fully cost-loaded and include the assignment of all financial resources to each scheduled activity. The Programme showing the erection portion of the Contract shall show the skilled resources but not the unskilled labour.

The Programme shall show the critical path and the float on non-critical activities. All design, manufacture, procurement, delivery, construction, erection and commissioning details shall be shown. All deliverables, access, Hold Points and Witness Points required by the Employer, completion, hand over and payment milestones shall be clearly marked. Histograms and progress curves may be requested by the Engineer.

The **fully cost-loaded and resource-assigned** Baseline Programme shall be agreed by the Contractor and the Engineer and shall be signed by the Contractor and the Engineer for purposes of identification and **for establishing the definitive performance measurement baseline (PMB). The contractor must submit, every seven days, an updated program for review to the engineer.**

Software Requirements

All Programmes shall be managed in a Microsoft Project 2003 compatible format and issued in a Microsoft Project 2003 compatible format.

Program Lay-out

The format of the Gantt chart shall be in blue bar form for all non-critical tasks and red for all critical tasks.

Each Programme shall have the Contractors' logo, the Project logo, the name of the Contractor, revision number, Project name and Contract description and date of update and print when issuing to the Engineer.

The name of the Contractor's planner shall be in the right hand corner.

The Engineer's name shall be indicated below the name of the Contractor's planners as follows.

Accepted by:

The columns in the Programme to be provided shall be as follows:

- Work break down reference number;
- Work area description;
- Planned percentage
- Actual percentage complete;
- Activity duration;
- Planned start;
- Planned finish;
- Actual start;
- Actual finish;
- Baseline start;
- Baseline finish.

Work shall be broken down into the appropriate lifecycles i.e. design, supply, install and commissioning. All Plant to be manufactured and installed shall be clearly indicated.

All deliveries, access dates, required permit dates, permit clearance dates, pressure tests dates, and synchronizing dates, hand over dates, ready for hot commissioning dates shall be marked as milestones.

Programmes shall be issued to the Engineer on A3 paper when issued in hard copy form, backed up by a copy in electronic format.

Programme Requirements

The Programmes shall be drawn up as per the Contractor's tender, but on a lower summarized level (Level 3). The Programmes baseline shall only be set once approved by the Project Manager. The Engineer will monitor the Contractor's progress against the baseline Programme.

Resourcing Requirements

All Site work shall be resourced. A manpower histogram may be requested by the Engineer prior to commencement of the on Site work.

Revisions to Programs

Each Programme issued to the Engineer shall have a revision number clearly indicated. Programmes shall only be revised if changes are made to the scope of work.

Formal notification of any date or scope changes made by the Employer that will impact on the Contractor will be communicated, then the Contractor shall revise its Programme accordingly and issue to the Employer for approval.

Progress Updating

Updated Programmes with the latest progress shall be issued to the Engineer once every 14 days before construction and 7 days (on every Friday) during construction.

Updated Programmes shall indicate:

1. actual percentage versus planned percentage;
2. duration to completion;
3. actual start and completion dates of tasks;
4. progress up to the Programme status date;
5. any tasks shown that should have started but were not so started shall be re-scheduled;
6. a summary showing milestones activities running late or ahead of schedule shall be attached to the Programme update when submitted to the Engineer.

Reporting

The progress report shall indicate the following:

1. all schedule implications;
2. highlight all possible deviations or problems and the corrective actions to be taken;
3. indicate issues that are causing delays or impeding progress;
4. updated S-curve shall accompany reports;

5. the S-curve shall indicate "planned" versus "actual" progress and forecast to completion curve based on current rate of progress;

6. a 2 (two) week look-ahead report shall be issued to the Engineer every two weeks by the Contractor.

Critical Milestone Dates

The following critical milestones shall, amongst others, be included in the Programme:

- Contract Commencement Date;
- Deliveries;
- Access requirements;
- Witness tests;
- Milestone payments;
- Taking-over – Time for Completion

Other Important notes:

- All Zambian registered companies are advised of the Bank of Zambia Act No 43 of 1966 that requires that the quotation be made in the Zambian unit of Currency, the Kwacha.
- Contractor to note that the contract value shall remain fixed during the duration of the contract.
- Contractor to note that tender maybe disqualified due to erasures and alterations.
- If any rework is done due to poor workmanship, the contractor will undertake this at his/her own cost. Including replacement of materials, if required.
- Invoicing shall only be done after successfully commissioning the system and completion certificates signed off.

Pricing

- All BOQs to be priced.

KEY DATES

#	**Project Task**	**Target Dates**
1	Contract Commencement Date	
2	Deliveries and Access requirements	
3	Witness tests	
4	Milestone payments	

5	Taking-over – Time for Completion	
6	Hand over dates / Completion Certificates signed off	
7	End of Defects Liability Period	

DOCUMENTS ATTACHED TO THIS SCOPE.

Title of Document	**Document Number**	**Revision**
Ablution block drawing		
Lower Secondary drawing		
Primary Block		
Upper Secondary		
Admin		

BOQ

Unabridged BOQ attached as **Appendix A, (TO BE COMPLETED)**

ZAMBIA SUGAR RESPONSIBILITIES

For safe access to the site, all Illovo site safety procedures must be followed with respect to the following:

- Issuing of PO to the contractor
- Approval of HSE File
- Approval of Method Statement, Risk Assessment and Quality Control Plan
- Permit to Work

TYPICAL CONTRACTOR AND CLIENT SCOPE OF SUPPLY

- Applicable to work done at Zambia Sugar Premises.
- In the event of load shedding, the Contractor must supply their own back up generator

Quality Assurance Hold Points

- A High level Quality Control Plan must be submitted for approval to the Engineer for commencement of work but at a bare minimum must include items in Table 1.

Table 1: Hold points

Project Phase	Quality Control Hold Point / Item	Type of Test / Inspection	Acceptance Criteria
I. Materials & Pre-Work	Material Samples Approval	Visual Inspection / Laboratory Test Certificates	Materials (e.g., tiles, locks, bulb holders) must be: Approved by the Engineer prior to procurement (as per Preamble vi).
	Structural Drawings	Document Review / Clash Detection	Drawings (including rebar and steel shop drawings) must be Approved by the Engineer before fabrication/fixing.
II. Substructure & Earthworks	Site Setting Out	Tape Measurement / Total Station Survey	Must conform to Approved Benchmarks and Coordinates (Dimensional Tolerance: typically ±10mm).
	Foundation Excavation	Tape Measure / Visual Inspection	Trenches must be excavated to Required Depths and the bottom must be firm, clean, and undisturbed (Hold Point before blinding concrete).
	Anti-Termite Treatment	Visual Inspection / Certificate of Application	Treatment must be applied as specified by the manufacturer to all soil surfaces before DPC/concrete placement.
	Backfilling & Compaction	Nuclear Density Gauge Test / Modified Proctor Test (Lab)	Material must be compacted in 150mm layers to achieve a minimum of 95% (Maximum Dry Density) MDD.

III. Structural Works	Formwork & Rebar Fixing	Tape Measure / Visual Check (Concrete Cover Blocks)	Reinforcement must be fixed as per approved shop drawings, with minimum 25mm concrete cover maintained (Hold Point before concrete pouring).
	Concrete Pour	Slump Test (Field) / Cube Crush Test (Lab)	Slump must be within the specified range (±25mm). Cube crush test must achieve minimum 25MPa at 28 days.
	Masonry / Blockwork	Visual Inspection / Dimensional Check	Blocks must be solid concrete with minimum strength of 7.5 N/mm² and laid true, plumb, and square using 1:4 cement-sand mortar.
IV. Finishes & Services	Plumbing Pressure Test	Hydrostatic Pressure Test (Gauge)	uPVC/PPR PN10 piping must be pressure tested to 1.5 times the system working pressure (or min. 6 bar) for one hour before concealment (No leak/pressure drop).
	Electrical First Fix	Continuity Test (Multimeter) / Visual Inspection	Conduits and wires must be installed as per layout and all circuits must demonstrate continuity and be visually compliant with approved drawings.
	Final Electrical System	Insulation Resistance Test / Earth Loop Impedance Test	System must be fully tested (e.g., insulation resistance > 1MΩ) and fully compliant with IEE Regulations (BS7671) and local standards prior to energization/handover.
V. Handover & Closeout	Final Inspection	Walk-through / Snagging List	The final construction must be free of defects (snags) and conform in all respects to the Approved Drawings and Specifications.

SAFETY

Apart from the general safety requirements already mentioned in ZAMBIA SUGAR contracts, the following specialized safety training is required:

- None

RISK RESPONSIBILITY

RISK	RESPONSIBLE PARTY
Delivery of materials to site	Contractor (within Contractor's scope)
Load shedding	Contractor
Mobile crane availability	Zambia Sugar
Project Overrun	Contractor
Poor workmanship / quality	Contractor

DECLARATION BY CONTRACTOR:

The Contractor will acknowledge that the contract will be withdrawn immediately, if Z.S finds the Contractor recruiting inside the Factory premises.

The rates inserted by the Tenderers will be deemed to include all requirements in these notes Labour, Materials and other free issue items, workmanship, Machinery and all ancillaries for completion of the contract.

By the submission of the Tender. Tenderers will be deemed to have fully acquainted themselves with the Tender document and to have thoroughly investigated the site and conditions, access to the local requirements and laws, supply of materials, matters affecting manufacture, delivery and storage and all aspects of construction, envisaged in the document prior to pricing and submission of their tender.

I/We have made myself / ourselves fully conversant with work to be done, have studied the tender document comprising (inter alia):

- Relevant Specifications.
- Zambia Sugar SHE Specifications
- Zambia Sugar Safety Rules and Regulations for Contractors.
- Any other Special Safety Requirements.

And understand that this tender, if accepted by us, will be deemed as a contract under the standard trading conditions of Zambia Sugar for the due completion of the work in the time and for the sum specified.

Contractor's signature…… Signature of witness……

Designation …… Designation ……

Date …… Date ……

Company address and stamp……